Current makes a magnetic field

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect current direction with the magnetic field around a wire.

SUCCESS CRITERIA - I can...

1. Sketch circular field lines around a straight wire.
2. Use a stated right-hand rule.
3. Predict the effect of reversing current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “magnetic field” to explain your model or result. Sentence starters are optional.

current: rate of charge flow

magnetic field: a field that influences magnets and moving charges

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

A 6 Ω resistor is in series with two parallel 8 Ω resistors across 20 V.
Find parallel equivalent, total resistance and supply current.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $B = \mu_0 I/(2\pi r)$; $\mu_0 = 4\pi \times 10^{-7}$ T·m/A.
Long straight wire. Right thumb follows conventional current; curled fingers show field direction.

Worked reasoning: Around a straight wire carrying conventional current out of the page, the magnetic field circles counterclockwise by the right-hand rule. A dot means out of page.

Your turn: Draw three field arrows around the wire and explain what the dot represents.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Current out of page (dot). Add field arrows.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Conventional current points out of the page.
Use the right-hand rule to draw the surrounding magnetic field direction.

Sentence starter: The … in my model represents … . Its label shows … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

04 Apply the physics

A long straight wire carries 5.0 A out of the page.
Calculate B at 0.020 m using $B = \mu_0 I/(2\pi r)$, $\mu_0 = 4\pi \times 10^{-7}$ T·m/A.
Mark the location and local field direction on the model.

Sentence starter: I found … [unit]. This value means … in this situation.

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

05 Explain your reasoning

What happens to field direction if the current reverses while its magnitude stays the same?

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A neutral object is attracted to an object carrying +4 nC.
Explain how attraction can occur without giving the neutral object a net charge.
Label charges and both interaction forces.

Word help: charge = a property responsible for electric interactions

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: 1 2 3

07 Exit check

Current triples and distance doubles. Express new B as a multiple of the old value.

Sentence starter: I notice … . This supports … because … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

For a long straight wire use $B=\mu_0 I/(2\pi r)$, with $\mu_0=4\pi\times10^{-7}$ T·m/A.
Find B for 5 A at 0.10 m; then double distance.

Sentence starter: I found … [unit]. This value means … in this situation.

Cross = current into page. Add magnetic-field arrows.



Marzano check - circle: 1 2 3